Technical Service Bulletin

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1999 ISM Engine INtroduction (T702)	

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

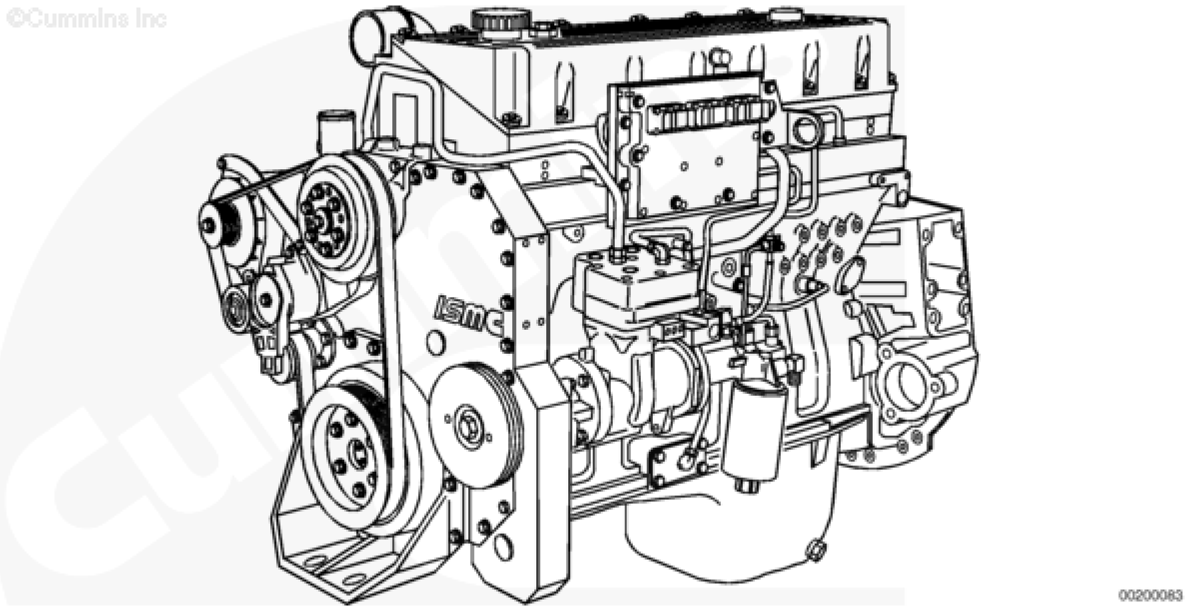
This service parts topic announces the introduction of the 1999 ISM engine. The ISM engine was designed to meet lower emission requirements.

Changes associated with the 1999 ISM engine include:

- Automatically tensioned fan drive
- Ratings
- Wastegated turbocharger
- Lubricating oil drain intervals

Automatically Tensioned Fan Drive

Beginning in 1999, automotive engines will offer an automatically tensioned fan drive. Reference the illustration below.



1999 ISM Engine with Automatically Tensioned Fan Drive

Main Components		
Quantity	Part No.	Part Name
1	3400881	Alternator support
1	3400884	Idler pulley
1	3400885	Tensioner
1	3400878	Alternator brace
1	3400880	Alternator pulley
1	3400877	Crankshaft pulley

Related Items			
Quantity	Part No.	Part Name	Description
1	3081348	Hex head capscrew	M10 X 90 (10.9)
2	3821847	Capt. wash. capscrew	M10 X 60 (9.8)
1	3009330	Washer	M10 - plain
1	3401204	Idler pulley cover	Fabricated
1	3400882	Idler pulley shaft	Fabricated
1	3820710	Capt. wash. capscrew	M10 X 80 (9.8)
1	3818060	Capt. wash. capscrew	M10 X 25 (9.8)
1	3009330	Washer	M10 - Plain
1	S-145	Hex head capscrew	½-13 X 1.25 in
1	3895516	Hex head capscrew	½-20 X 6.00 in

Related Items			
1	70897	Locknut	1/2/20
1	3035806	Washer	½ in
1	3895517	Washer	½ in
1	3896043	Clamping plate	0.035-in thick
5	3896790	Hex head capscrew	M14 X 63 (10.9)

V-Ribbed Belts					
Quantity	Part No.	No. of Ribs	Effective Length	Fan Center	Drive Ratio
1	3099255	8	69.25 ± 0.22 in	304.8 mm [12 in]	1.2:1
1	3099256	8	70.50 ± 0.22 in	304.8 mm [12 in]	1:1
1	3099256	8	70.50 ± 0.22 in	330.2 mm [13 in]	1.2:1
1	3099257	8	73 ± 0.22 in	381 mm [15 in]	1.2:1
1	3099258	8	75.50 ± 0.22 in	431.8 mm [17 in]	1.2:1
1	3099259	8	78.50 ± 0.22 in	482.6 mm [19 in]	1.2:1
1	3099260	8	71.75 ± 0.22 in	330.2 mm [13 in]	1:1
1	3099261	8	83.50 ± 0.22 in	533.4 mm [21 in]	1:1
1	3099290	8	81.75 ± 0.22 in	533.4 mm [21 in]	1.2:1
1	3099291	8	80.25 ± 0.22 in	482.6 mm [19 in]	1:1

Ratings

The 1999 wastegated and nonwastegated turbocharged engines are rated as follows:

Wastegated Turbocharged Engines			
Application	CPL	Rated Horsepower	Advertised Torque
Linehaul	2350	425 @ 1800 rpm	1550 @ 1200

Wastegated Turbocharged Engines			
Vocational	2350	425 @ 2100 rpm	1550 @ 1200
Vocational	2350	425 @ 2100 rpm V	1450 @ 1200
Linehaul (Top 2)	2350	425 @ 1800 rpm ST1	1450/1550 @ 1200
Linehaul (Top 2)	2350	380 @ 1800 rpm ST1	1450/1550 @ 1200
Fire Truck	2350	500 @ 1800 rpm	1550 @ 1200
RV	2350	500 @ 1800 rpm	1550 @ 1200

Nonwastegated Turbocharged Engines			
Application	CPL	Rated Horsepower	Advertised Torque
Fire Truck	2608	450 @ 1800	1450 @ 1200
Fire Truck	2608	400 @ 1800 rpm	1350 @ 1200
Fire Truck	2608	400 @ 1800 rpm	1450 @ 1200
Fire Truck	2608	350 @ 2100 rpm	1350 @ 1200
Fire Truck	2608	370 @ 2100 rpm	1350 @ 1200
Fire Truck	2608	330 @ 2100 rpm	1350 @ 1200
Fire Truck	2607	350 @ 2100 rpm	1200 @ 1200
Fire Truck	2607	320 @ 2100 rpm	985 @ 1200
RV/Vocational	2608	450 @ 1800 rpm	1450 @ 1200
Linehaul	2608	410 @ 1800 rpm	1450 @ 1200
Vocational	2608	400 @ 1800 rpm	1450 @ 1200
Linehaul	2608	350 @ 1800 rpm	1350 @ 1200
Vocational	2608	350 @ 2100 rpm	1350 @ 1200
ESP	2608	350/400 @ 1800 rpm	1350 @ 1200
Linehaul	2608	370 @ 1800 rpm	1350 @ 1200
Vocational	2608	370 @ 1800 rpm	1350 @ 1200
ESP	2608	370/410 @ 1800 RPM	1350 @ 1200
Linehaul	2608	370 @ 1800 rpm	1450 @ 1200
Vocational	2608	370 @ 2100 rpm	1450 @ 1200
Linehaul	2608	330 @ 1800 rpm	1350 @ 1200
Vocational	2608	335 @ 2100 rpm V	1450 @ 1200
Linehaul	2607	310 @ 1800 rpm	1150 @ 1200
Linehaul	2607	330 @ 1800 rpm	1250 @ 1200
ESP	2607	330/370 @ 1800 rpm	1250 @ 1200
Linehaul	2607	280 @ 1800 rpm	1050 @ 1200
Vocational	2607	280 @ 2100 rpm	1050 @ 1200

Nonwastegated Turbocharged Engines			
Vocational	2607	310 @ 1800 rpm ST1	1150 @ 1200
Linehaul (Top 2)	2607	310 @ 1800 rpm ST1	1150/1250 @ 1200
Linehaul (Top 2)	2607	330 @ 1800 rpm ST1	1250/1350 @ 1200
Linehaul (Top 2)	2608	330 @ 1800 rpm ST1	1350/1450 @ 1200
Linehaul (Top 2)	2608	370 @ 1800 rpm ST1	1350/1450 @ 1200
Linehault (Top 2)	2608	400 @ 1800 rpm ST1	1350/1450 @ 1200
Urban Bus	2201	330 @ 2100 rpm	1040 @ 1200
Urban Bus	2201	280 @ 2100 rpm	900 @ 1200

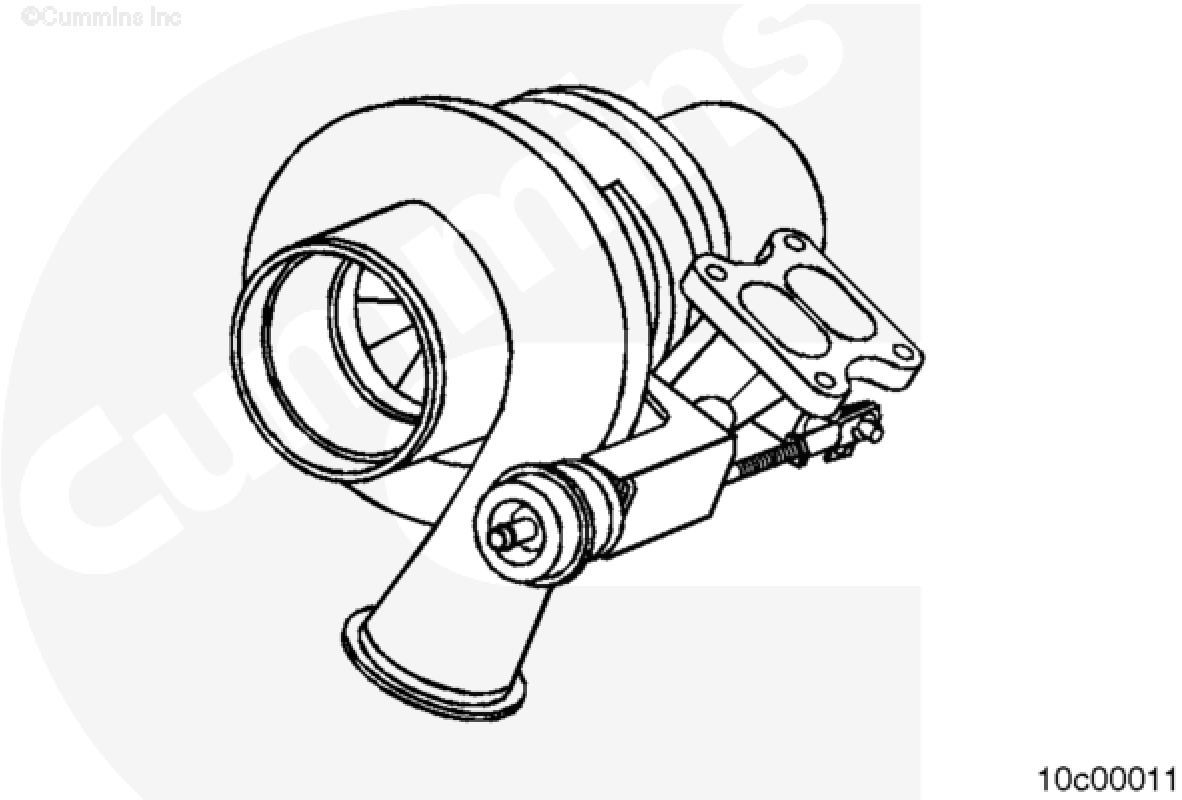
Australian Ratings			
Application	CPL	Rated Horsepower	Advertised Torque
Linehaul	2477	330 @ 1800 rpm	1250 @ 1200
Linehaul	2477	330 @ 1800 rpm	1350 @ 1200
Linehaul	2477	350 @ 1800 rpm	1350 @ 1200
Vocational	2477	350 @ 2100 rpm	1350 @ 1200
Linehaul	2477	370 @ 2100	1350 @ 1200
Vocational	2477	370 @ 2100 rpm	1350 @ 1200
ESP	2477	370/410 @ 1800 rpm	1350 @ 1200
Linehaul	2477	370 @ 1800 rpm	1450 @ 1200
Vocational	2477	370 @ 2100 rpm	1450 @ 1200
Linehaul	2477	400 @ 1800 rpm	1450 @ 1200
Vocational	2477	400 @ 1800 rpm	1450 @ 1200
*Linehaul	2610	425 @ 1800 rpm	1550 @ 1200
*Vocational	2610	425 @ 2100 rpm	1550 @ 1200
*Wastegated Turbocharged engine.			

Wastegate Turbocharger

The wastegate turbocharger is a model HX55 with improved wheel clearances for optimized performance. It is comprised of a turgocharger, wastegate actuator, and wastegate valve in the turbine housing.

The wastegated turbocharger was designed to provide improved response at low engine speeds without sacrificing turbocharger durability at high speeds. This is accomplished by allowing precisely measured exhaust gases to bypass the turbine wheel during certain modes of engine operation.

During low-rpm operation, the turbocharger operates as a closed-system turbocharger and all of the exhaust gases flow through the turbine side of the turbocharger where the gases' heat energy is transferred to the compressor wheel and used to compress intake air. During high-rpm operation, however, the turbocharger becomes an open-system turbocharger and allows exhaust gas to bypass the turbine. Since exhaust gas is gated around the turbine wheel, less energy is absorbed through the turbine and transferred to the compressor, allowing for precisely controlled intake manifold pressures and turbine speeds.

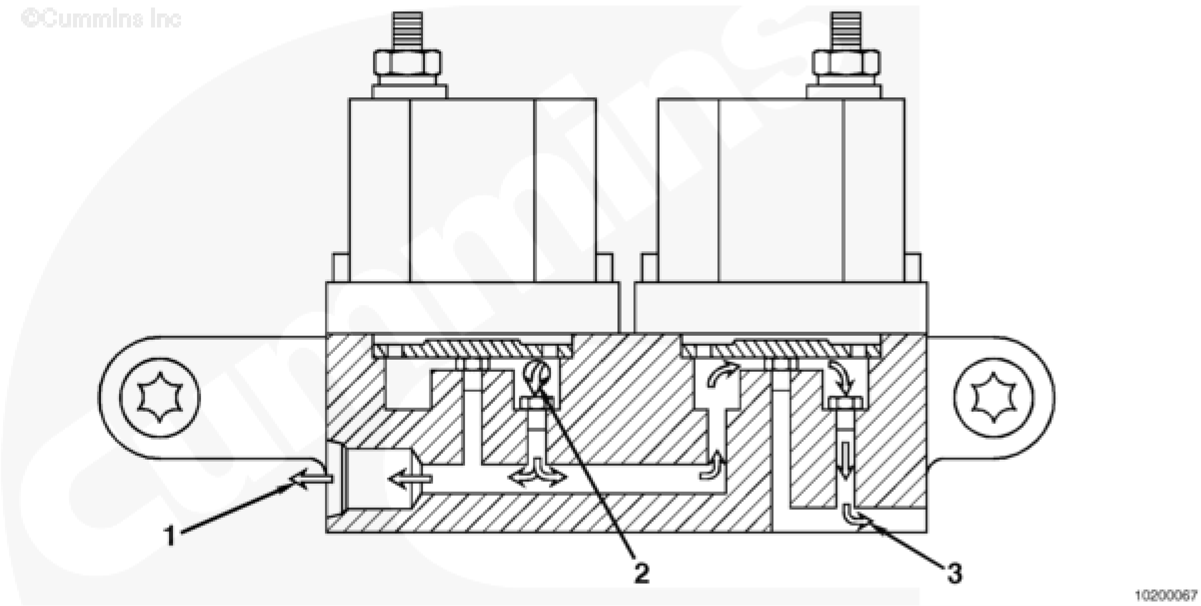


Wastegated Turbocharger Assembly

Wastegate Controller

The wastegate controller is mounted on the rear of the intake manifold on the turbocharger side of the engine and is controlled by the ECM. The wastegate controller regulates the percentage of the intake manifold pressures sent to the wastegate actuator. Two solenoids, similar to the fuel shutoff valves on the CELECT™ product, are used to regulate this percentage.

Part No.	**Description**
3348688	12-VDC Controller Assembly
3348686	24-VDC Controller Assembly



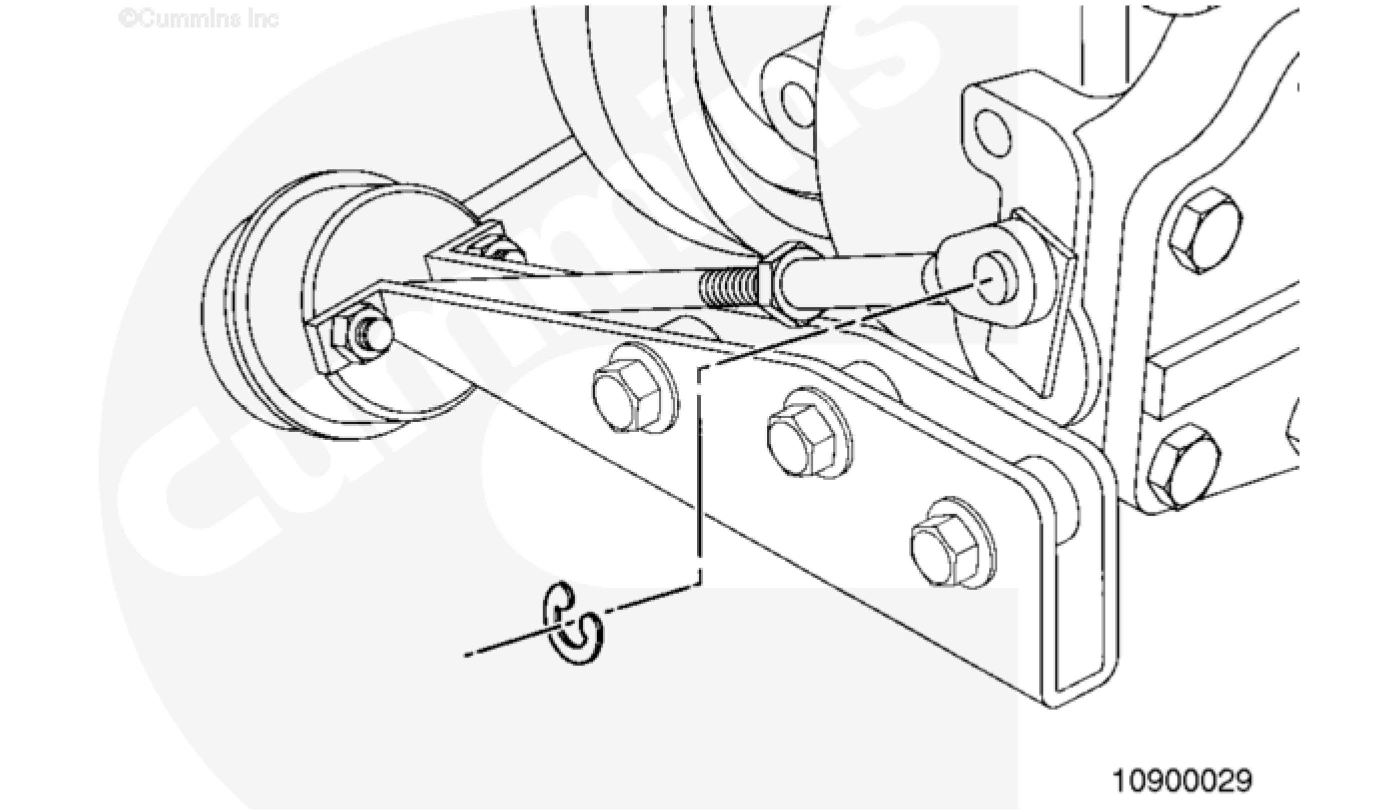
Wastegate Controller: (1) to Wastegate Actuator, (2) from Intake Manifold, (3) to Atmosphere

Wastegate Actuator

The wastegate actuator is mounted on the turbocharger and consists of a pressure canister, diaphragm, and rod. As pressure increases in the canister, as dictated by the wastegate controller, the actuator rod opens the wastegate vale.

Wastegate Valve

The wastegate valve is mounted inside the turbocharger in the turbine housing. As the valve opens, exhaust gas is allowed to bypass the turbine wheel, lowering turbine speed to optimize intake manifold pressure.



Wastegated Turbocharger Actuator and Valve Assembly

Lubricating Oil-Drain Intervals

Is the vehicle one of those below?

- Linehaul truck
- Coach bus
- Equipment that accumulates 8000 miles a month or more

IF YES:

Select the correct oil-drain interval from the following tables based on engine turbocharger, engine duty cycle, and oil type.

Maximum Oil-Drain Intervals

- Follow oil-drain interval Severe if the vehicle operates under either of the conditions listed in interval Severe.
- Follow oil drain interval Normal if the vehicle operates under either of the conditions listed in interval Normal and does not meet any conditions in interval Severe.
- Follow oil-drain interval Light if vehicle operates under both of the conditions listed in interval Light and does not meet any conditions in interval Severe or interval Normal

Table 1				
Nonwastegated Turbocharged Engine Oil-Drain Interval by Severity of Service km [mi]				
Oil Class	Severe	Normal	Light	Acceptable Oil Filters

Table 1				
Fuel Burned - km/l [mpg] Gross Vehicle Weight	< 2.6 [6.0] 80,000	2.6 [6.0] to 3.0 [7.0] 70,000 to 80,000	3.0 [7.0] < 70,000	
API CG-4	16,000 [10,000]	24,000 [15,000]	40,000 [25,000]	LF9001
CES 20071 (CH-4)(1)	24,000 [15,000]	48,500 [30,000]	64,500 [40,000]	LF90001
CES 20076(2)(3)	32,000 [20,000]	56,327 [35,000]	72,500 [45,000]	LF9001
Notes:				
1. API CH-4 can be used as an alternative to CES 20071.				
2. If the engine is equipped with Centinel™, lubricating oil with API grade meeting Cummins Engineering Standard (CES) 20076 is required to maintain engine performance and durability. Oil-drain intervals using Centinel™ will be 525,000 viles with oil-filter change at 75,000 miles. The use of a high-quality filter is mandatory. Use Cummins Part No. 3406809 or Fleetguard® Part No. LF9001. These criteria apply equally to automotive and nonautomotive engines.				
Valvoline® Premium Blue® and Premium Blue® 2000 meet CES 20076 standards.				

Table 2				
Wastegated Turbocharged Engine Oil-Drain Interval by Severity of Service km [mi]				
Oil Class	Severe	Normal	Light	Acceptable Oil Filters
Fuel Burned - km/l [mpg] Gross Vehicle Weight	< 2.6 [6.0] 80,000	2.6 [6.0] to 3.0 [7.0] 70,000 to 80,000	3.0 [7.0] <70,000	
API CG-4	8000 [5000]	13,000 [8000]	19,500 [12,000]	LF9001
CES 20071 (CH-4)(1)	16,000 [10,000]	24,000 [15,000]	40,000 [25,000]	LF9001
CES 20076 (2)(3)	20,000 [12,500]	32,000 [20,000]	48,500 [30,000]	LF9001
Notes:				
1. API CH-4 can be used as an alternative to CES 20071.				
2. If the engine is equipped with Centinel™, lubricaating oil with API grade CES 20076 is required to maintain engine performance and durability. Oil-drain intervals using Centinel™ will be 525,000 miles with oil-filter change at 75,000 miles. The use of a high-quality filter is mandatory. Use Cummins Part No. 3406809 or Fleetguard® Part No. LF9001. These criteria apply equally to automotive and nonautomotive engines.				
Valvoline® Premium Blue® and Premium Blue® 2000 meet CES 20076 standards.				

IF NO:

Select the correct drain interval from the following list based on engine horsepower rating and the oil type used from Tables 3, 4, and 5.

Table 3 Nonlinehaul Oil-Drain Interval				
Nonwastegated Turbocharged Engines				
Oil Class	km	Miles	Hours	Months
CG-4	7000	4500	200	6
CES 20071 (CH-4) [(1)]	11,500	7000	300	6
CES 20076 [(2)]	14,500	9000	400	6
Wastegated Turbocharged Engines				
Oil Class	km	Miles	Hours	Months
CG-4	5500	3500	150	6
CES 20071 (CH-4) [(1)]	7000	4500	200	6
CES 20076 [(2)]	11,500	7000	300	6
Notes:				
1. API CH-4 can be used as an alternative to CES 20071.				
2. Valvoline® Premium Blue® and Premium Blue® 2000 meet CES 20076 standards.				

Table 4				
Specialty, RV, and Fire Truck Ratings - 450 hp				
Oil Class	km	Miles	Hours	Months
CG-4	7000	4500	200	6
CES 20071 (CH-4)[(1)]	11,500	7000	300	6
CES 20076[(2)]	14,500	9000	400	6
Specialty, RV, and Fire Truck Ratings - 500 hp				
Oil Class	km	Miles	Hours	Months
CG-4	4000	2500	100	6
CES 20071 (CH-4)[(1)]	5500	3500	200	6
CES 20076[(2)]	7000	4500	200	6
Notes:				
1. API CH-4 can be used as an alternative to CES 20071.				
Valvoline® Premium Blue® and Premium Blue® 2000 meet CES 20076 standaards.				

Please follow these oil-drain intervals for transit-bus applications.

Table 5				
Maximum Oil-Drain Intervals by Average mph				
	CG-4 Engine Oil	CH-4 or CES 20071 Engine Oil	CES 20076 [1] Engine Oil	Acceptable Oil Filter
Avg. km/h [mi]	km [mi]	km [mi]	km [mi]	
26 to 29 [16 to 18]	6500 [4000]	13,000 [8000]	15,000 [9500]	LF9001
23 to 26 [15 to 16]	5500 [3500]	11,000 [7000]	13,500 [8500]	LF9001
19 to 23 [12 to 14]	5000 [3500]	9500 [6000]	11,000 [7000]	LF9001
16 to 19 [10 to 12]	4000 [2500]	8000 [5000]	9500 [6000]	LF9001
12 to 16 [8 to 10]	3000 [2000]	6500 [4000]	8000 [5000]	LF9001
10 to 12 [6 to 8]	2400 [1500]	5000 [3000]	5500 [3500]	LF9001
6 to 10 [4 to 6]	1500 [1000]	3000 [2000]	4000 [2500]	LF9001
Note:				
1. Valvoline® Premium Blue® and Premium Blue® 2000 meet CES 20076 standards.				

Lubricating Oil Filter

The wastegated and nonwastegated turbocharged engines offer the use of the LF9001 oil-lubricating filter.

Note : The LF9000 filter is not defined for use with the 1999 ISM wastegated or nonwastegated turbocharged engine.

Note : Do not replace the LF9001 filter with the extended length LF9000 filter without verifying adequate clearance with suspension and/or chassis at extremes of travel to avoid possible engine damage. Oil- and Filter- change intervals can not be extended if the LF9000 oil filter is used in place of the LF9001.

Document History

Date	Details
2011-1-17	Module Created

Last Modified: 21-Jan-2011